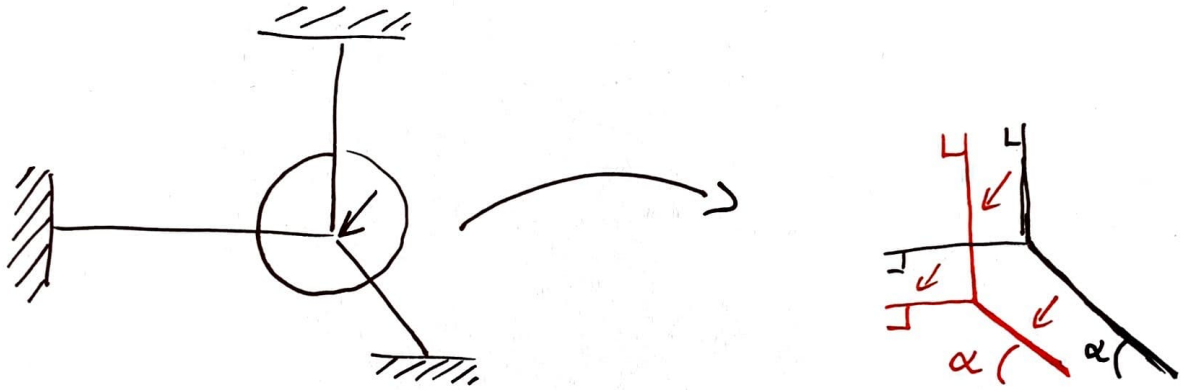
Övning 3 - Statiskt obestämda stångbärverk

2.2: 4, 11, 14, 15

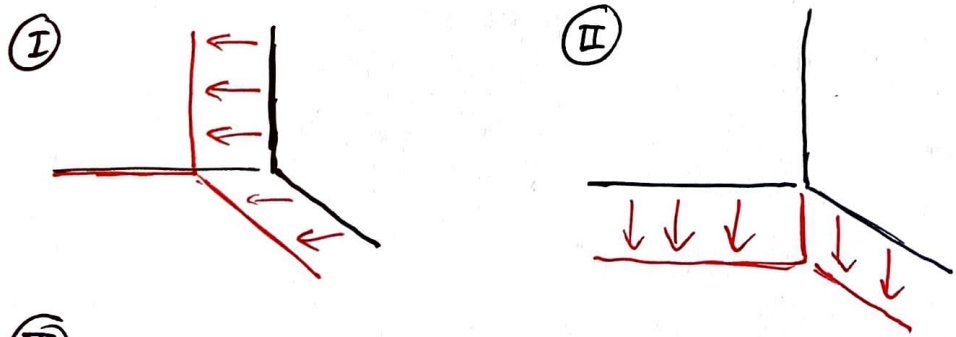
Deformationssamband

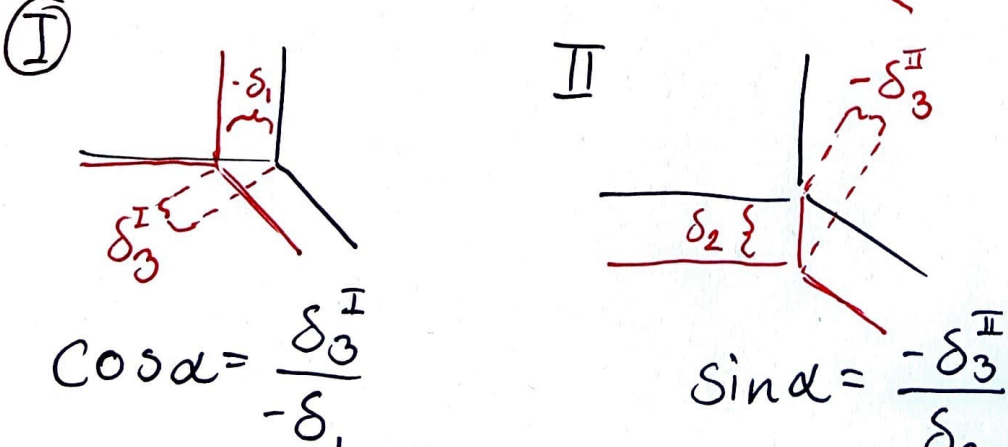
- Små deformationer
  - Alla vinklar bevaras



Knutpunkten rör sig, men alla vinklar bevaras!

Dela upp deformationen





$$\cos\alpha = \frac{\delta_3^I}{-\delta_1}$$

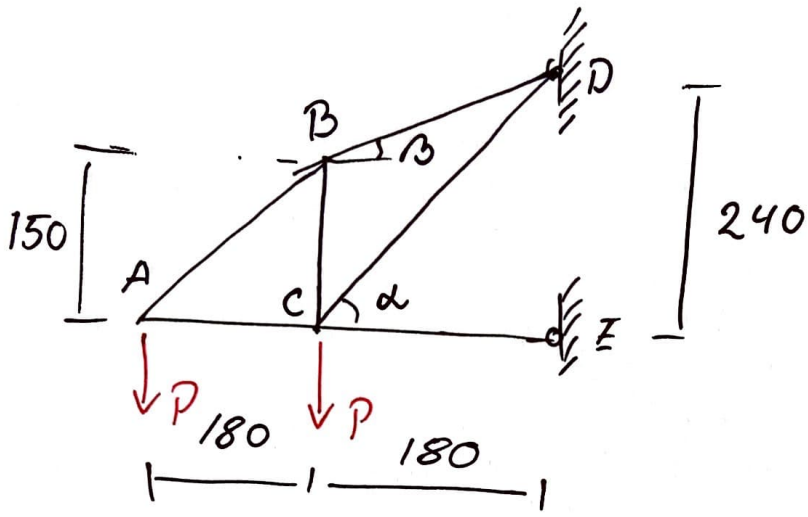
$$\sin\alpha = \frac{-\delta_3^{II}}{\delta_2}$$

$$\Rightarrow \boxed{\delta_3 = \delta_3^I + \delta_3^{II} = -\delta_2\cos\alpha - \delta_1\sin\alpha}$$

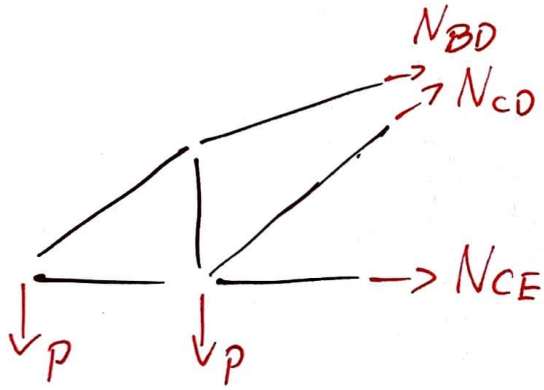
2.2.4

Sökt: Normalspänningen i stång CD

$|N_{CD}|$



① Snitta och ta jämvikt



$\uparrow: -2P + N_{BD}\sin\beta + N_{CD}\sin\alpha = 0$

$\rightarrow: N_{CE} + N_{BD}\cos\beta + N_{CD}\cos\alpha = 0$

$\circlearrowleft: +180P + 150\,N_{BD}\cos\beta = 0$

ekv 3 ger $N_{BD} = \dfrac{180P}{150\cos\beta}$

insatt i ekv 1

$$N_{CD}\sin\alpha = 2P - \frac{180P}{150\cos\beta}\sin\beta$$

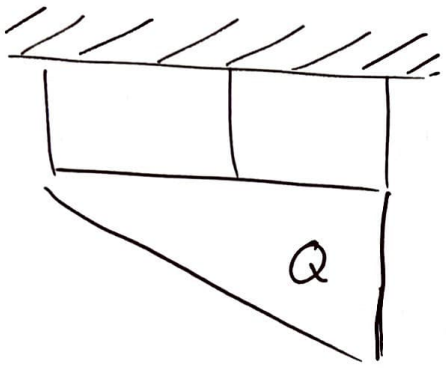
$$\sin\alpha = \frac{240}{\sqrt{240^2+180^2}} = 0.8 = \frac{4}{5}$$

$$\tan\beta = \frac{90}{180} = \frac{1}{2}$$

$$N_{CD} = \frac{5P}{2} - \frac{5}{4}\frac{180}{150}\cdot\frac{1}{2}P = \frac{7}{4}P$$

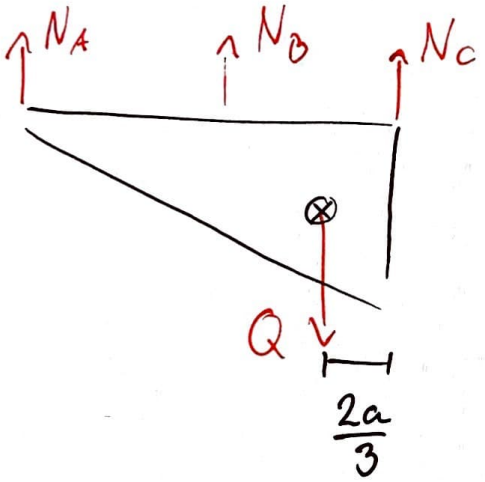
$|N_{CD} = 1.75P|$

2.2.11



Sökt: Normalkrafterna i alla stänger
$|N_A \; N_B \; N_C|$

① Snitta och ta jämvikt



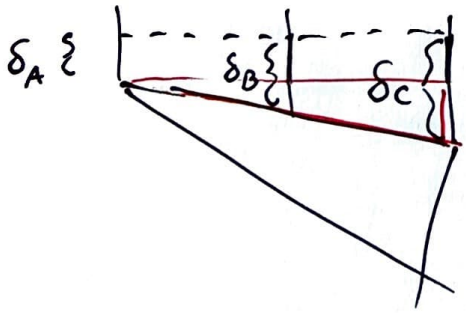
Masscentrum fås från Beta s. 237

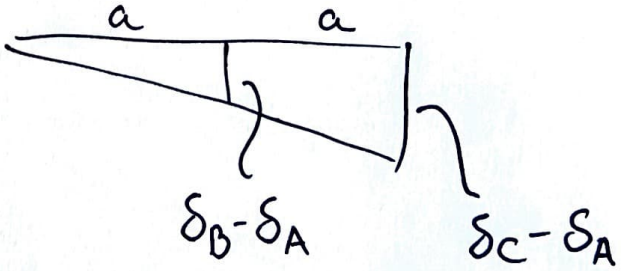
② $\uparrow$: $N_A + N_B + N_C = Q$

$\circlearrowleft C$: $\frac{2a}{3}Q - aN_B - 2aN_a = 0$

3 okända
2 ekvationer

③ Kompatibilitetssamband





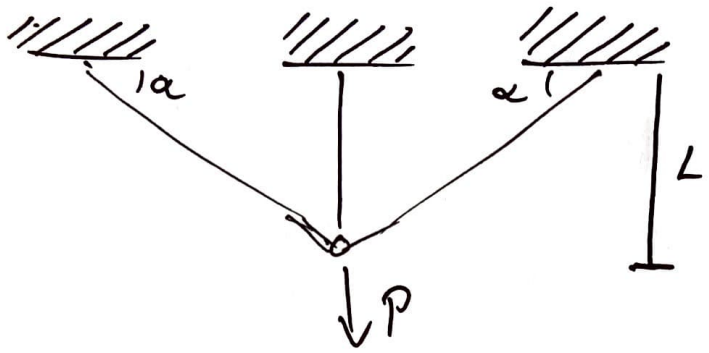
$$\frac{\delta_B - \delta_A}{a} = \frac{\delta_C - \delta_A}{2a} \Rightarrow 2\delta_B = \delta_C + \delta_A \Rightarrow \delta_B = \frac{\delta_C + \delta_A}{2}$$

$$2N_B = N_C + N_A \Rightarrow \boxed{N_B = \tfrac{1}{3}Q \quad N_A = \tfrac{1}{6}Q \quad N_C = \tfrac{1}{2}Q}$$

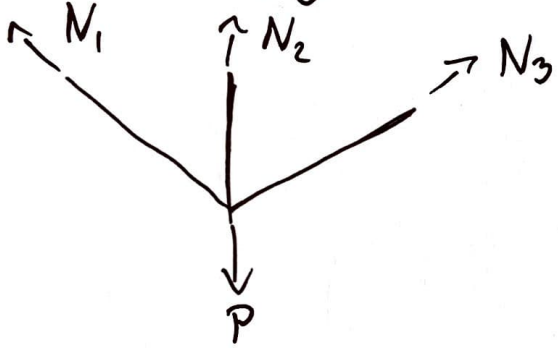
2.2.14

Givet: A, E, α, L

Sökt: Normalkrafterna i stängerna N_1 N_2 N_3



① Frilägg



$\uparrow: N_2 + N_1 \sin\alpha + N_3 \sin\alpha - P = 0$

$\rightarrow: N_3 \cos\alpha - N_1 \cos\alpha = 0$

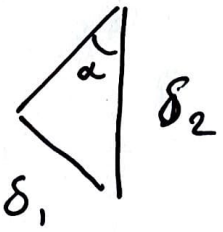
$\Rightarrow N_1 = N_3$ (symmetri!)

$N_2 + 2N_1 \sin\alpha = P$ (I)

2 okända 1 ekv

② Deformationssamband





$\sin\alpha = \frac{\delta_1}{\delta_2}$ (II)

$\delta_i = \frac{N_i L_i}{EA}$ $\quad \delta_1 = \frac{N_1 \frac{L}{\sin\alpha}}{EA}$ $\quad L_1$ $\quad \delta_2 = \frac{N_2 L}{EA}$

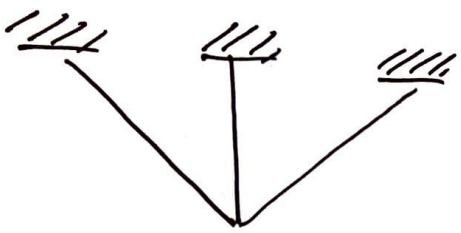
(II) $\Rightarrow \frac{N_2 L}{EA} = \frac{N_1 L}{EA} \frac{1}{\sin^2\alpha} \Rightarrow N_2 = N_1 \frac{1}{\sin^2\alpha}$

(I) $N_1 \frac{1}{\sin^2\alpha} + 2N_1 \sin\alpha = P \Rightarrow N_1 = \frac{P \sin^2\alpha}{2\sin^3\alpha + 1}$

$N_2 = \frac{P}{2\sin^3\alpha + 1}$

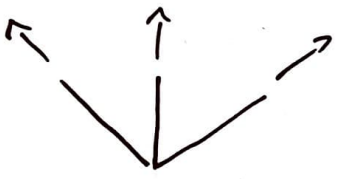
2.2.15

Sökt: Normalkrafterna $N_1\ N_2\ N_3$ ③



Passningsfel, Stängarna dras ner Δ för att sedan fjädra tillbaka

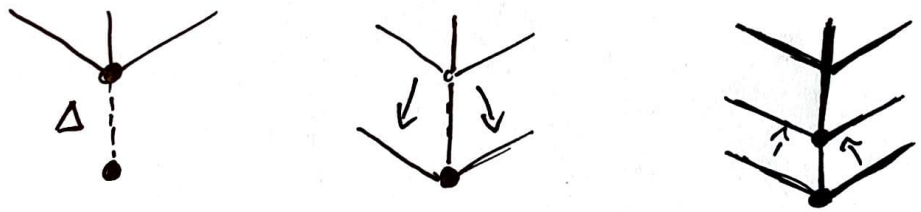
① Snittning och jämvikt

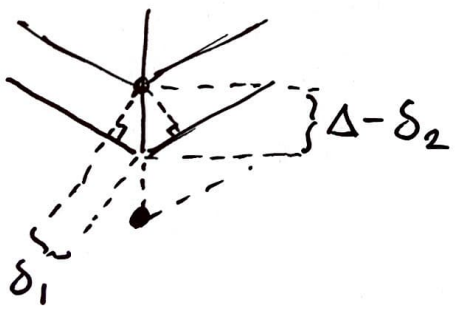


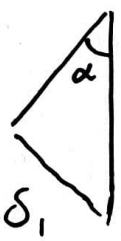
$\rightarrow$: $N_1 = N_3$

$\uparrow$: $2N_1 \sin\alpha + N_2 = 0$ (I)

③ Deformationssamband







$\sin\alpha = \dfrac{\delta_1}{\Delta - \delta_2}$ (II)

$\delta_1 = \dfrac{N_1 L}{EA\sin\alpha}$ $\quad$ $\delta_2 = \dfrac{N_2 L}{EA}$

(II) $\left(\Delta - \dfrac{N_2 L}{EA}\right)\sin\alpha = \dfrac{N_1 L}{EA\sin\alpha} \Rightarrow N_1 = \left(\dfrac{EA\Delta}{L} - N_2\right)\sin^2\alpha$

(I) $2\left(\dfrac{EA\Delta}{L} - N_2\right)\sin^3\alpha + N_2 = 0$

$$N_2 = -2\,\frac{EA\sin^3\alpha\,\Delta}{L(1+2\sin^3\alpha)}$$

$$N_1 = \frac{EA\sin^2\alpha\,\Delta}{L(1+2\sin^3\alpha)} = N_3$$